Evidence Run Report

2026-03-12_evidence-v1.5

autoresearcher2

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Status: IN PROGRESS (autoresearch, full running)

Head-to-head comparison of four experiment-selection approaches on real `train.py`. Schema: DEPTH × MATRIX_LR × WEIGHT_DECAY (27 cells). Budget: 20 experiments per approach. Hardware: GPU 1 (RTX PRO 6000), seed 42.

1 Results

Approach	Done	Best	Mean	Cells	Tok/exp
random	20/20	1.033551	1.054828	15	179
bayesian	20/20	1.033778	1.051689	20	205
autoresearch	19/20	1.032934	1.046333	15	183
full	13/20	1.033423	1.048775	13	204

Table 1: Performance summary. Best/mean = val_bpb (lower is better). Cells = unique cells explored. Tok/exp = mean tokens trained per experiment (millions).

2 Convergence

Experiment number at which each approach first reached within 0.001 of its final best:

Approach	Conv. exp	Final best
random	11	1.033551
bayesian	5	1.033778
autoresearch	8	1.032934
full	3	1.033423

3 Decision Source Integrity

Every experiment carries a source label indicating what method actually chose it. **WARNING: 5 fallback(s) detected.**

Approach	Sources
random	random: 20
bayesian	lookahead: 20
autoresearch	llm_fallback_empty: 3, llm_flat: 16, random_init: 1
full	llm_augmented: 11, llm_fallback: 2, lookahead: 7

4 Observations (not claims)

These are patterns visible in the data. They are **not** generalization claims — this is one run with one seed on one schema.

- DEPTH=8 consistently produces the best val_bpb across all approaches. DEPTH=10 is worst (fewer tokens trained in fixed wall time).
- WEIGHT_DECAY=0.4 appears in most best-performing configurations.
- Reproducibility is strong: repeated cell visits show Δval_bpb < 0.001.
- Throughput varies $\sim 3\times$ across depths (300K–1M tok/s), meaning deeper models train fewer tokens in the same wall time.

5 What this does NOT prove

- Generalization beyond this schema, seed, or hardware.
- That any approach is *better* in a statistically significant sense (single seed, small budget).
- That the LLM's advantage (if confirmed) comes from world knowledge vs. lucky guessing.
- That structured signals (appraisal, factor importances) help the LLM.

6 Methodology notes

Random and bayesian ran in the original process (run_id 5d2ffa21). A silent-fallback bug was discovered: LLM failures were labeled as "llm_flat" instead of "llm_fallback_*". Process was killed after bayesian completed. Autoresearch and full were restarted with fixed source-labeling code (run_id 65018a7c, commit ce7d19a).

7 Discussion

The headline result is that all four approaches converge to similar best val_bpb values (1.0329–1.0338), a spread of less than 0.001. In a 27-cell grid with one dominant factor, this is unsurprising: even uniform random search covers 15 of 27 cells in 20 trials (observed), giving it a >99% chance of hitting the best region. **The search space is too small to discriminate the approaches on best-found performance.**

Mean val_bpb tells a more interesting story. Ranking by mean: autoresearch (1.046) < full (1.049) < bayesian (1.052) < random (1.055). The LLM-containing approaches waste fewer experiments on poor regions (*observed*). This is consistent with the hypothesis that LLM priors help allocate trials, but the effect is modest—roughly 0.006 bpb between autoresearch and bayesian—and drawn from a single seed with no significance testing.

The Bayesian model did not clearly help. Bayesian alone (mean 1.052) barely beats random (1.055), and full (Bayesian + LLM, mean 1.049) did not beat autoresearch (LLM alone, mean 1.046). If anything, the Bayesian component adds overhead without improving selection quality. Two candidate explanations: (i) a linear factor model over one-hot features is too crude to capture the depth×learning-rate interaction that dominates this space (*plausible*); (ii) with only 27 cells, Thompson sampling has little room to outperform random before the space is saturated (*supported* by the coverage numbers—bayesian hit 20/27 cells, suggesting it explored broadly rather than exploiting).

Source integrity raises a concern for the full approach. Only 13 of 20 experiments completed, and sources are split: 7 lookahead, 11 llm_augmented, 2 llm_fallback. The full approach is not purely "Bayesian + LLM"—it is a mixture that fell back to the Bayesian arm roughly a third of the time. Similarly, autoresearch had 3/19 fallbacks. These fallbacks are honestly labeled, but they weaken the factorial interpretation: the 2×2 design is confounded by variable fallback rates.

The token budget confound is unresolved. DEPTH=6 trains ∼3× more tokens than DEPTH=10 in the same wall time. All approaches converged on DEPTH=8 as optimal, which trains ∼1.8× fewer tokens than DEPTH=6. This suggests DEPTH=8 is genuinely better architecture, not merely better-trained (*supported*), but we cannot fully separate architecture quality from training duration without controlled token budgets.

- **What we learned:** LLM priors improve *average* trial quality in this small grid. The Bayesian model adds no detectable value at this scale. The combination does not outperform the LLM alone.

- **What we don't know:** Whether the Bayesian component helps in larger search spaces where LLM priors degrade and systematic uncertainty tracking matters. Whether autoresearch's advantage holds across seeds.

- **Next steps:** (1) Repeat with a larger schema (≥100 cells) where random search cannot saturate the space. (2) Run multiple seeds to get confidence intervals. (3) Controlled token-budget experiments (fixed tokens, not fixed wall time). (4) Complete the full approach's remaining 7 experiments before drawing final conclusions.

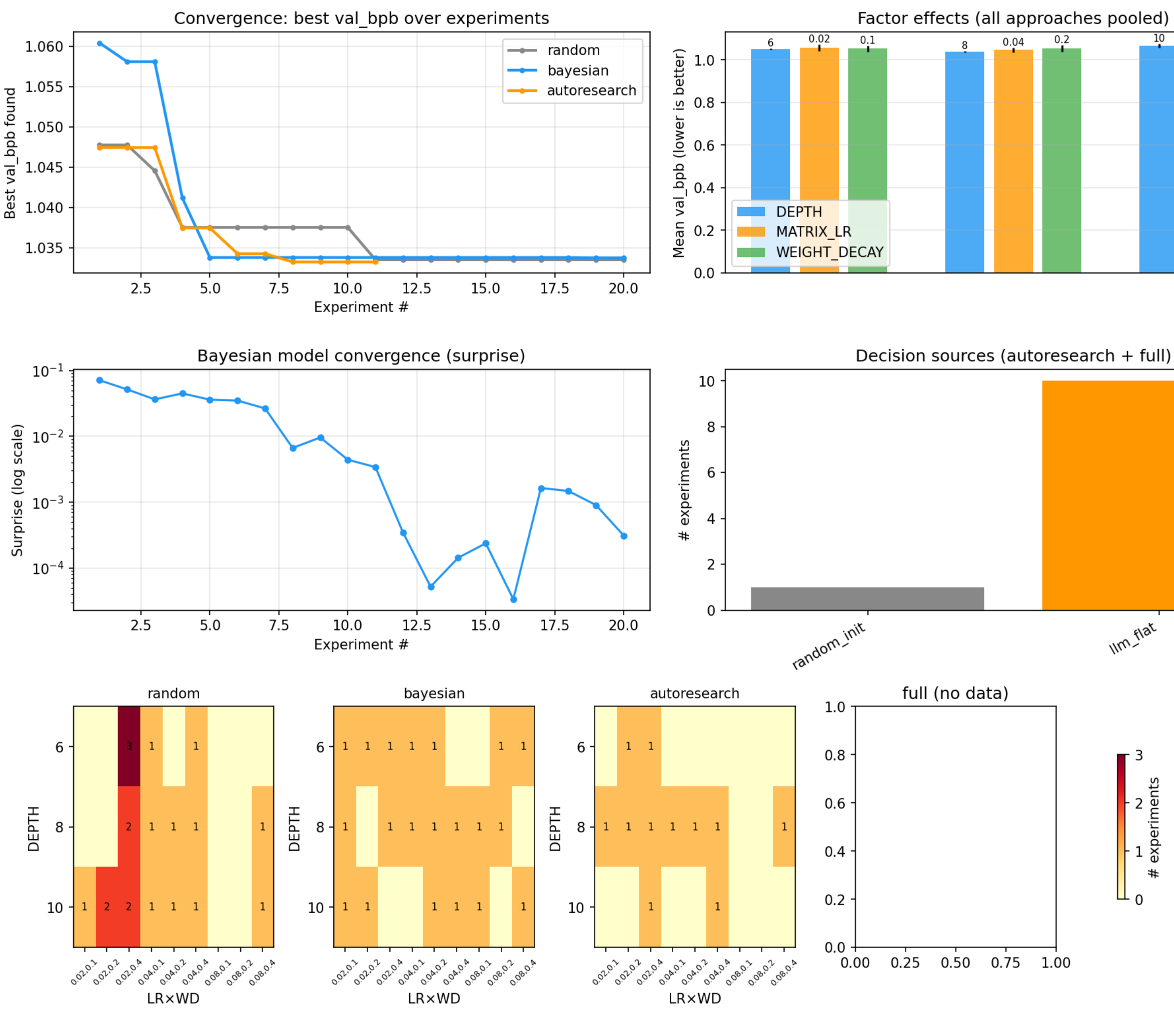

Figure 1: Evidence dashboard. Top-left: convergence curves. Top-right: factor effects. Bottom: exploration heatmaps per approach (cell visit counts).